PII detection in logs

-P2 -

P2

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Abstract:

Here is the abstract

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Preface

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Use of AI
In this report, generative AI such as ChatGPT, Gemini, and Copilot have been used throughout the project as guidance. While this project is a collaboration between multiple parties, AI has been useful in combining different formalities and writing styles to get a more uniform language. Furthermore, AI tools are used to support the making of figures, visualizations, and models. This usage has had a positive impact on both the learning processes and learning outcomes. It is used as a supplement, not a replacement for critical reflection or academic work.

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Acronyms

AI Artificial Intelligence. 7, 11, 16

API Application programming interfaces. 12

BERT Bidirectional Encoder Representations from Transformers. 13

CCPA California Consumer Privacy Act. 4

CDC Change Data Capture. 12

CRF Conditional Random Fields. 13, 26–30

CSV Comma-separated values. 9

EU European Union. 4, 5

GDPR General Data Protection Regulation. 1, 4–7, 14, 16–18

GPT Generative Pre-trained Transformer. 14

IAPP International Association of Privacy Professionals. 3

IDF Inverse Document Frequency. 32

L-BFGS Limited-memory Broyden–Fletcher–Goldfarb–Shanno. 29

LLM Large Language Model. 9

LSTM Long Short-Term Memory. 14

ML Machine Learning. 7, 11, 12, 15, 26

NER Named Entity Recognition. 8, 11, 13, 15

NLP Natural language processing. 8, 15

PII Personally identifiable information. 1–4, 7–19, 23, 25, 31, 32

SVM Support Vector Machine. 21, 30, 31

TF Term Frequency. 32

TF-IDF Term Frequency-Inverse Document Frequency. 32

UCPA Utah Consumer Privacy Act. 4

Chapter 1

Introduction

The digitalisation of society is continuously increasing, and the collection and processing of personal data have become central to how companies and organizations operate worldwide. Large databases containing customer information are used to optimize services, improve products, and support data-driven decision-making. At the same time, the growing reliance on personal data raises legal and ethical challenges, as innovation and individuals' rights to privacy and data protection must be balanced.
Modern IT systems continuously generate large volumes of logging data, which may contain Personally identifiable information (PII), making it important to handle and protect this data appropriately to prevent misuse. Due to the scale and complexity of this data, manual detection is often insufficient. This creates a need for automated solutions capable of efficiently identifying and classifying sensitive information.

This project is given by the master data consultancy *Combine*, which helps other companies with digital optimization. Therefore, this project investigates how Machine Learning can be applied to detect PII within log data and automatically identify and classify sensitive information. By exploring these technological approaches, the project seeks to minimize the risk of PII being stored in unintended logs or exposed in ways that violate data protection laws, such as General Data Protection Regulation (GDPR).

1.0.1 Initial problem statement

How can Machine Learning be utilized to detect PII so it does not end up in incorrect logs?

Chapter 2

Problem analysis

2.1 PII

PII is the abbreviation of "Personally Identifiable Information" and refers to all information linked to an individual, either directly or indirectly. This data can be collected when you purchase something online, register a car, visit a doctor, or engage in many other activities in which personal information is shared with a company, organization, or other parties.
PII can take various forms, such as text, images, sound, or video, as long as the information can be used to identify a specific person. [1] [2]

2.1.1 Personal Data Categories

Within the context of PII, data is often categorized based on its level of sensitivity and linkability. Linkable data includes items such as driver's license numbers, passport numbers, and Social Security numbers. In contrast, non-linkable data includes information such as first names, ZIP codes, or age, and may require additional information to identify an individual on its own. However, when combined with other data, it can be linked and cause harm.
Both types are therefore considered PII and can be further classified as either General PII or Sensitive PII. [3][2]

Category	**Examples**
General PII	Name, Email, Age, Phone number, Address
Sensitive PII	Social security number, Passport number, Credit card number, Bank accounts, Religion, Political opinions, Sexuality, Ethnic background

Table 2.1: Categories of PII [3][2]

As shown in Table 2.1, the first category consists of General PII. Although this information is considered less sensitive, it is still PII and must be protected.
The second category consists of sensitive PII. This type of information requires higher protection because it may pose a significant risk if shared or misused. Therefore, there are strict laws on the collection, storage, and processing of this type of data.[3] [2]

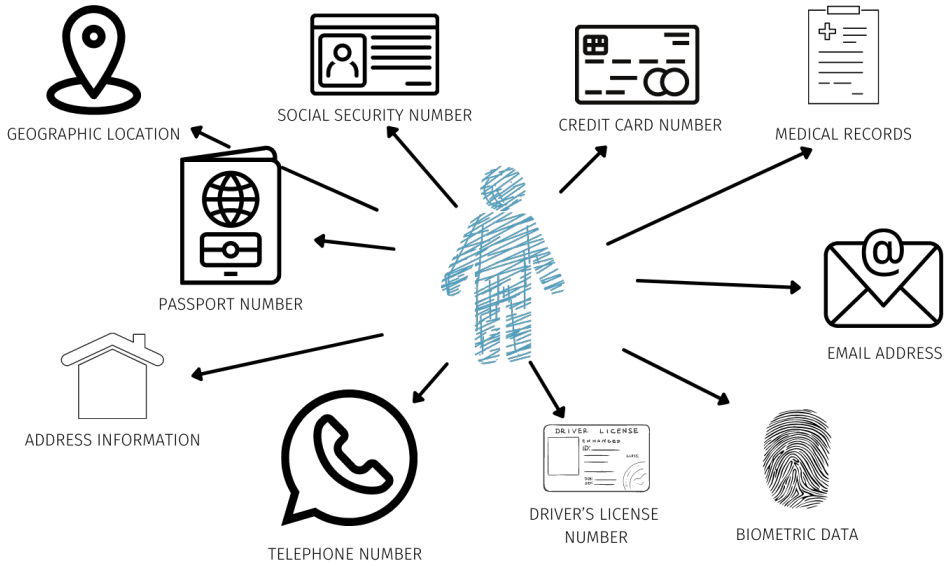

Figure 2.1: Illustration of different categories of Personally Identifiable Information (PII) [4]

Figure 2.1 illustrates examples of PII linked to an individual. The person placed at the center represents the private person, while the surrounding elements show different types of personal data that can be connected to that individual. It includes both general and sensitive PII. It demonstrates that, when combined, multiple data points can clearly identify a person and increase the potential for privacy risks. [4]

Understanding what PII is and the sensitivity of the data is essential for companies and organizations, as it defines the scope of data that must be handled in accordance with data protection laws. [2]

2.1.2 Importance of Protecting PII

It is important to protect PII, especially from misuse, data breaches, and cyberattacks, and this is critical for maintaining trust between users and organizations.
According to the International Association of Privacy Professionals (IAPP), approximately

68% of global users are concerned about their online privacy, and this concern influences their behavior when engaging with companies and digital services [5]. Organizations that fail to implement robust privacy protections risk losing customers, as users may choose to avoid or disengage from services where they perceive a risk to their personal data. Therefore, ensuring proper data protection not only fulfills legal requirements but also plays a key role in strengthening consumer trust and confidence in digital systems. [3]

2.1.3 PII Laws and Regulations Worldwide

Approximately 75% of countries worldwide have data privacy laws, which means that PII regulations differ from country to country. However, most of these follow a similar distinction between general and sensitive data [3].
As an example, data protection laws in the United States consist of multiple state-level laws, creating a complex patchwork without a unified national privacy law [6].
Some examples include the California Consumer Privacy Act (CCPA), which gives California residents the right to access their personal data and request deletion, and the Utah Consumer Privacy Act (UCPA), which gives Utah residents the right to know what personal data businesses collect about them. Additionally, the federal Privacy Act of 1974 Controls how personal information is held by federal agencies, is kept, used, and accessed. [3] [6]

Although regulations vary by jurisdiction, all handling of PII across multiple regions is often required to comply with several sets of rules. In practice, this often makes companies adapt to the strictest standards, which is the European law GDPR, across all operations to simplify compliance and reduce legal risk [3].

2.2 GDPR

GDPR is considered one of the world's strictest security laws and the most well-known data regulation law. It came into use on May 25, 2018, and was introduced because older data protection systems, such as the European Data Protection Directive from 1995, could not keep up with modern technology and the fast changes in internet usage and online data storage. [7]
GDPR is a set of data protection laws in the European Union (EU) that apply when handling personal data. This typically includes situations where companies process information about their customers or employees. The purpose of GDPR is to strengthen individuals' rights and give them greater control over their own personal data. [8]

2.2.1 Data Collection and Minimization

It can be tempting for organizations to collect as much data as possible. However, GDPR enforces the principle of data minimization, which ensures that only data necessary for a specific purpose is collected and processed. Personal data must only be collected when there is a clear reason, and collecting irrelevant information is not allowed.
This principle forces companies to focus on the essential data needed to achieve their intended purpose, while minimizing unnecessary data to reduce the risk of data breaches and misuse. Data minimization also requires that unnecessary data be deleted or anonymized if it happens to be collected. Although data minimization does not prevent all risks of data leakage, it reduces the potential harm to individuals by limiting the amount of personal information that could be exposed. [9]

2.2.2 Roles and Responsibilities

An essential part of the GDPR is the concept of data processing. Data processing encompasses any action performed on personal data, such as collection, registration, organization, storage, use, sharing, or deletion. The regulation ensures that all these activities are carried out securely and lawfully.[10] The GDPR applies to all companies, organizations, and public authorities within the EU, but it also extends to organizations outside the EU if they process personal data of individuals in the EU. This means that non-EU companies must comply with the GDPR when handling data from EU residents. The regulation further defines distinct roles in data processing as seen in Table 2.2. [10]

Role	**Description**
Data Subject	The individual to whom the personal data belongs. The GDPR protects the data subject's privacy and rights, including the right to access their data, correct inaccurate data, and request deletion of their data and the right to be forgotten.
Data Controller	The person, company, or organization that determines the purpose and means of processing personal data. The data controller is legally responsible for ensuring that personal data is processed in compliance with the GDPR.
Data Processor	The person or organization that processes personal data on behalf of the data controller. The data processor must follow the controller's instructions and ensure appropriate security measures are in place to protect the data.

Table 2.2: GDPR Roles and Responsibilities [10]

Furthermore, figure 2.2 illustrates the relationship between the data subject, data controller, and data processor. The data subject provides personal data to the controller, who decides why and how the data should be processed. The processor carries out the processing strictly according to the controller's instructions. The figure emphasizes that the controller holds primary responsibility for GDPR compliance, while the processor acts under delegated authority. [10] [11]

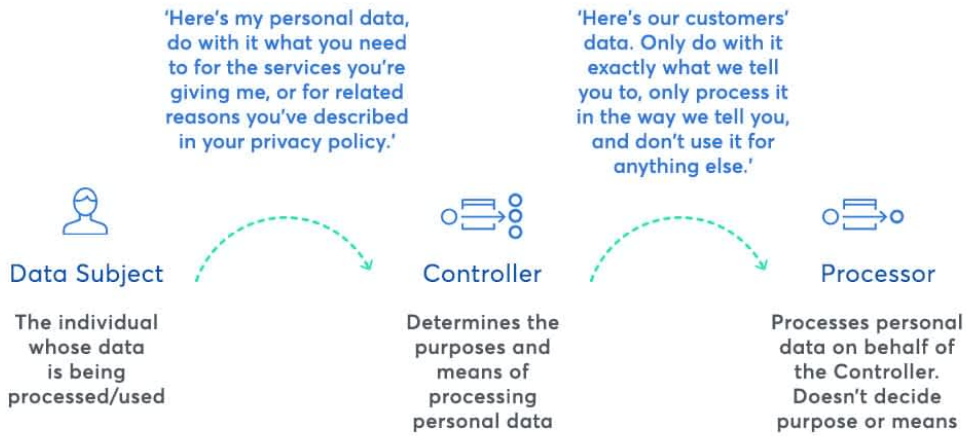

Figure 2.2: Data flow and role distribution between Data Subject, Controller, and Processor[11]

2.2.3 Compliance and Consequences

It is important to comply with the GDPR because violations can result in significant fines. Supervisory authorities are responsible for monitoring compliance and can impose penalties for breaches of the regulation. Fines can be up to 10 million euros or 2% of a company's annual global turnover for less severe violations. For serious violations, fines can reach up to 20 million euros or 4% of the company's annual global turnover. [10]

2.3 Dataset

The chosen English dataset is designed for the detection and classification of PII. Hugging Face[12] is a collaborative platform for the Artificial Intelligence (AI) and Machine Learning (ML) community. It is a place for developers and researchers to host, share, and collaborate on models, datasets, and demo applications.

The creator of the dataset is a Danish consultancy named "Syv.ai", and is based in Copenhagen. As a small but focused and ambitious team, they specialize in bringing AI into the everyday operations of Danish businesses[13].
They specialize in a deep understanding of both technology and business, and help their clients harness the full potential of AI. Syv.ai translates complex AI solutions into practical tools that enhance efficiency and innovation in clients' daily operations[13].

2.3.1 Difference in Data

The dataset consists of 266.000 data rows with annotated personal information[14]. It includes different categories of PII where some of the most common types of data are names, email addresses, phone numbers, physical addresses, credit card numbers, dates of birth, and usernames. These data types correspond to commonly regulated personal data categories under data protection legislation, including GDPR [15].

One important characteristic of the dataset is the heterogeneity of its PII entities. The annotated data does not consist of a single homogeneous data type, but instead includes a mixture of textual, numeric, and alphanumeric information. This distinction has significant implications for model design and detection strategy.

Some PII categories are primarily text-based, such as personal names, street names, cities, and usernames. These entities are highly context-dependent. For example, a personal name may only be identifiable as PII when appearing in a specific syntactic or semantic context. Detecting such entities typically requires machine learning-based approaches, as purely rule-based methods are generally insufficient for reliably identifying contextual PII, since names may overlap with common nouns[16].
Other PII categories are primarily numeric, such as credit card numbers, phone numbers, dates of birth, and identification numbers. These types often follow specific structural patterns (e.g., fixed length, digit-only sequences, checksum rules). Numeric PII tends to be less context-dependent but more sensitive to formatting and structural patterns.

Some PII types combine letters and numbers, such as email addresses, user IDs, and account numbers. These entities typically follow semi-structured patterns. The presence of heterogeneous data types implies that a single detection strategy may not be optimal. Instead, the dataset naturally supports a multi-layer detection architecture, for example:

Rule-based detection for structured numeric PII, and Machine learning-based Named Entity Recognition (NER) for contextual textual PII [17].

This distinction is particularly relevant in real-time systems, as numeric pattern matching is computationally lightweight and suitable for low-latency environments, whereas transformer-based Natural language processing (NLP) models may introduce higher processing costs[18].

In the table 2.3 below are some examples of the data structure, selected at random from the chosen dataset. These examples demonstrate the different PII types and the text containing the PII within the dataset. The dataset is split into two columns: the Source Text and Labels. The source text, shown on the left, contains the PII. On the right, it shows the corresponding labels and their values, listed in a dictionary format.

Previous Therapist: Winkelhagen, Contact: +127.465.343.7176, Email: ingwinkelhprotonmail.com, Previous Job Area: IT, Current Residence: Town Line Avenue, ID Card Number: 1433931887, Previous Policy Number: US.EN.8592.G829	"label": "LAST_NAME", "value": "Winkelhagen" "label": "PHONE_NUMBER", "value": "+127.465.343.7176" "label": "EMAIL", "value": "ingwinkelh@protonmail.com" "label": "STREET", "value": "Town Line Avenue" "label": "ID_CARD_NUMBER", "value": "1433931887"
User profile updates needed by 19/05/2058. Verify 28HA, Haylee. Encrypt XPG9N5P626CYP04D7 and 938114. Ensure compliance with British Pound Sterling regulations. Contact +449-94.214.0864 for further details at Stair Road, TN10.	"label":"USERNAME", "value":"28HA" "label":"FIRST_NAME", "value":"Haylee" "label":"PHONE_NUMBER", "value":"+449 94.214.0864" "label":"STREET", "value":"Stair Road" "label":"ZIPCODE", "value":"TN10"
Investment Portfolio Report: Client: Ilmije Dekegel Account Number: 009561448205692 Balance: 118,795.78 City: Consett Your investments in the tech sector performed well. For details, contact us at D@yahoo.com.	"label": "FIRST_NAME", "value": "Ilmije" "label": "LAST_NAME", "value": "Dekegel" "label": "ACCOUNT_NUMBER", "value": "009561448205692" "label": "CITY", "value": "Consett" "label": "EMAIL", "value": "D@yahoo.com"
Applicant Info: Mx Faridin Fennouri Vehicle Plate No: QDG-580 Driver's License: TH7L5XUAM74NA Residential Address: North San Pedro Road, 94903	"label": "FIRST_NAME", "value": "Faridin" "label": "LAST_NAME", "value": "Fennouri" "label": "DRIVER_LICENSE_NUMBER", "value": "TH7L5XUAM74NA" "label": "STREET", "value": "North San Pedro Road" "label": "ZIPCODE", "value": "94903"

Table 2.3: Examples of the data structure from the dataset

2.3.2 Alternative Datasets

While the dataset shown earlier was provided by Combine, in combination with the purpose of this project. There exists alternative datasets which serves the same purpose, in being used as training data for PII detection systems. Here two examples will be taken at random from three different datasets, used for PII training. The examples will exclude some columns, as this sections only purpose is to show the formatting of the data not the data as a whole. The example datasets listed below was created using synthetic data:

Ai4Privacy Dataset

This dataset was created by "Ai4Privacy". Their goal is to prevent the potential risk of PII's being integrated into data pipelines, it consisting of over 200.000 different data points. It's written in a json format and contains 54 different types of sensitive PII data, in differing use cases such as formal documents, casual conversation, email's etc. Below are two examples showing the formatting of this dataset. [19]

source_text	target_text	privacy_mask	[...]	id	set
Dear Bette, we've noticed low engagement from your IP address 39de:ae2f:8ccf:d733:d79c: bc3c:a731:f26a. If you're facing network issues, please report them at the earliest.	Dear [FIRSTNAME], we've noticed low engagement from your IP address [IPV6]. If you're facing network issues, please report them at the earliest.	["value": "Bette","start": 5, "end": 10, "label": "FIRSTNAME","value": "39de:ae2f:8ccf:d733:d79c: bc3c:a731:f26a", "start": 62, "end": 101, "label": "IPV6"]	[...]	165,828	train
Creating a new strategy for intellectual property management. The meeting will be held in Basel-Landschaft, 40290, Von Mountain. Kindly verify your attendance.	Creating a new strategy for intellectual property management. The meeting will be held in [STATE], [BUILDINGNUMBER], [STREET]. Kindly verify your attendance.	["value": "Basel-Landschaft", "start": 90, "end": 106, "label": "STATE", "value": "40290", "start": 108, "end": 113, "label": "BUILDINGNUMBER", "value": "Von Mountain", "start": 115, "end": 127, "label": "STREET"]	[...]	166,366	train

Table 2.4: two examples from the dataset created by AI4Privacy, excluding: span_labels, mbert_text_tokens, mbert_bio_labels, language.

Kaggle PII dataset

Another usable dataset comes from Kaggle, created by Alejo Paullier. This dataset contains only 4434 Large Language Model (LLM) generated text examples containing PII. Unlike the dataset offered by Combine this is written in a Comma-separated values (CSV) file format. While similar in use case, this dataset has already had some preprocessing done, such as providing the text, tokens, labels and their individual values separated into their own columns. [20]

text	tokens	labels	name	phone	[...]	address	hobby
My name is Aaliyah Popova, and I am a jeweler with 13 years of experience. I remember a very unique ...	['My', 'name', 'is', 'Aaliyah', 'Popova,', 'and', 'I', 'am', 'a', 'jeweler', 'with', '13', 'years', ...	['O', 'O', 'O', 'B-NAME_STUDENT', 'I-NAME_STUDENT', 'O', 'O', 'O', 'O', 'O', 'O', 'O', 'O', 'O', 'O'...	Aaliyah Popova	(95)94215-7906	[...]	97 Lincoln Street	Podcasting
In the realm of aviation, where precision and safety intertwine, my name is Marceau Becker, an air t...	['In', 'the', 'realm', 'of', 'aviation,', 'where', 'precision', 'and', 'safety', 'intertwine,', 'my'...	['O', 'O', 'O', 'O', 'O', 'O', 'O', 'O', 'O', 'O', 'O', 'O', 'O', 'O', 'B-NAME_STUDENT', 'I-NAME_STUDENT'...	Marceau Becker	(681) 788-4947	[...]	9515 West 74th Avenue	Water skiing

Table 2.5: Dataset created by Alejo Paullier from kaggle, excluding: document, email, url, username, trailing_whitespace, prompt, prompt_id, job, len

PII in Financial Documents

Unlike the other two sets, this dataset is written in a financial context, created by Mendeley Data, this dataset was created for the purpose of detection PII in specifically financial documents. The dataset made use of realistic financial document structure to create it's fully synthetic data. It was created to resemble Tax filings, compliance notices and transaction confirmations, all of which would contain high amounts of sensitive PII. [21]

Name	[...]	SSN	Text	True Predictions
Kayla Griffin	[...]	840-01-1523	To: Kayla Griffin SSN: 840-01-1523 TIN: 840-01-1523 Address: 13481 Decker Ferry Suite 705 Kristina mouth, MD 42208 Subject: Confirmation of Tax Filing for FY 2023 Dear Kayla Griffin, Your tax filing for the fiscal[...]	[(4, 17, name), (189, 202, name), (772, 829, credit_card), (983, 1006, email), (1106, 1132, url), (1022, 1036, phone), (73, 125, address), (497, 522, company), (27, 38, ssn), (48, 59, ssn), (341, 352, ssn)]
James Hughes	[...]	165-55-7384	We are excited to welcome James Hughes to Montes Inc. As part of your onboarding, we have registered your personal details, including your home address (93796 Benjamin Junctions Woodwardchester, RI 87414), contact number [...]	[(26, 38, 'name'), (327, 384, 'credit_card'), (246, 263, 'email'), (465, 487, 'url'), (222, 236, 'phone'), (153, 203, 'address'), (42, 52, 'company'), (275, 286, 'ssn')]

Table 2.6: Two examples from Financial-Document by Mendeley Data, excluded data: Credit Card, Email, URL, Phone, Address, Copmany

Conclusion

While the alternative datasets shown above are all viable as training data for PII detection programs, this project will ultimately use the original dataset provided by Combine. It not

only encompasses many subjects in the curriculum, but also serves as a learning experience working with a company-provided database.

2.3.3 Synthetic Data

Working with synthetic data in this PII detection project has both methodological and practical implications that are directly relevant to AI, Data Systems, and ML [22]. Because the dataset is synthetic, it allows the development and evaluation of models without exposing real personal information. This supports ethical experimentation and reduces legal risk during the development phase. As demonstrated by the SPY dataset project[22], synthetic data offers a controlled environment with consistent annotations, proving highly advantageous for supervised PII detection tasks. It enables systematic comparison of models and clear evaluation using metrics such as precision, recall, and F1-score. The structured nature of the data makes it well-suited for training and benchmarking various detection approaches.

At the same time, synthetic data may differ from real-world enterprise data in important ways. Artificially generated examples are often cleaner, more structured, and less noisy than production data. In real systems, PII may appear in incomplete records, inconsistent formats, or mixed with technical metadata. This creates a potential domain gap between training and deployment. A model trained exclusively on synthetic data may perform well on test data but exhibit reduced robustness to unexpected input patterns. Therefore, it is important to critically reflect on generalization and not assume that high performance on synthetic data automatically transfers to real operational environments[22].

The heterogeneity of PII types in the dataset further influences model design. Text-based entities require contextual understanding and benefit from ML approaches such as NER, while structured numeric or semi-structured identifiers can often be detected using pattern-based methods [23]. This naturally supports a hybrid detection strategy, where lightweight rule-based components handle format-sensitive identifiers and ML models handle context-dependent entities. Such architectural considerations are particularly relevant in real-time systems, where computational efficiency and latency constraints must be balanced against detection accuracy.

In conclusion, synthetic data provides a safe and methodologically strong foundation for model development and experimentation in this project. However, it also requires a critical discussion of generalization, robustness, and system-level performance to ensure that the developed solution remains effective beyond the controlled training environment[22].

2.3.4 Scope

The processing of PII in software systems must distinguish between calls and logs. An API call or service invocation is a real-time interaction where PII may be transiently processed and filtered before storage. In contrast, a log is a persistent record that can grow sensitive information widely across systems and outside normal privacy governance controls. Logs containing raw PII become compliance liabilities, as they are often mirrored to monitoring tools, backups, and analytics - increasing the attack surface and potential for unauthorized exposure[24]. Research shows that even disciplined logging requires careful anonymization or masking to prevent privacy violations and regulatory breaches[25].

In this project, logs are used as a central component for system monitoring and documentation, but not as a primary data source for model training. Logs function as the system's "trace," recording events, processing steps, and potential errors throughout the data pipeline. When the system receives a data payload (e.g., via an Application programming interfaces (API) call or a Change Data Capture (CDC) event), the content is first processed by a PII detection component. The purpose is to identify and potentially mask or flag personally identifiable information before the data proceeds further in the system. Only after this processing step may relevant metadata be written to logs.

2.4 State of the art

For PII detection, different modeling approaches have been used over time. The preferred methods have evolved from rule-based systems to classical ML models, and more recently to deep learning architectures. Today, hybrid approaches combining rule-based methods, classical ML, and deep learning models are widely used in practice.

2.4.1 PII detection models

Rule-based
Among the earliest approaches are rule-based systems, where predefined patterns or regular expressions are used to identify PII. These systems perform well on structured data, where the format is known in advance (e.g., credit card numbers or social security numbers). However, they are sensitive to variation. For example, if a user omits a dash (-) in a credit card number or slightly changes the format, the rule-based system may fail to detect the information[16].

Machine-learning
Classic machine learning models have been applied to PII detection as a more flexible alternative to purely rule-based approaches. These models are trained on labeled datasets, enabling them to generalize from known examples and classify previously unseen data.

Unlike rule-based systems, classical Machine Learning models can learn statistical patterns from the data rather than relying solely on predefined rules. However, they still share certain limitations with rule-based methods. In particular, their ability to recognize patterns is constrained by the quality and representativeness of the training data and the chosen feature representation. As a result, they may fail to detect unseen variants or novel forms of PII, especially when such patterns deviate significantly from those observed during training [23].

Among supervised Machine Learning models, Conditional Random Fields (CRF) are considered more flexible than traditional independent classifiers such as Naive Bayes or logistic regression. Unlike these models, which classify each instance independently, CRF operate on sequences of data and model dependencies between neighboring labels[17].

A CRF assigns labels to each element in a sequence (e.g., tokens in a sentence) based on both the observed data and the labels of adjacent elements. By modeling these sequential dependencies, CRFs can capture contextual information that independent classifiers cannot. This allows the model to consider how previous labels influence the current prediction, making it particularly suitable for structured prediction tasks such as NER and PII detection[17].

Because CRF incorporate contextual and sequential information, they often outperform traditional classical Machine Learning models and purely rule-based systems in tasks where entity boundaries and label consistency are important. However, their performance still depends on the quality of feature engineering and the representativeness of the training data [17].

Deep learning

The modern approach to PII detection increasingly relies on neural network architectures, particularly deep learning models. These models can operate in various ways, including sequence modeling and contextual representation learning. Unlike traditional Machine Learning models that depend heavily on manually engineered features, neural networks automatically learn complex patterns and representations from large amounts of data.

Neural models are capable of capturing a broader range of features, including semantic, syntactic, and contextual information. By leveraging contextual embeddings, they can evaluate each token not only based on its own characteristics but also in relation to surrounding words and the broader linguistic context. This enables them to detect more subtle or previously unseen variations of PII[26].

In recent years, transformer-based architectures such as Bidirectional Encoder Represen-

tations from Transformers (BERT) and Generative Pre-trained Transformer (GPT) have become dominant in natural language processing tasks, including Named Entity Recognition and PII detection. These models rely on self-attention mechanisms that allow them to model long-range dependencies within text more effectively than earlier recurrent architectures such as Long Short-Term Memory (LSTM). Through large-scale pretraining and subsequent fine-tuning, transformer models can leverage contextual embeddings to improve generalization across diverse domains and linguistic variations[27].

However, despite their strong performance, neural network models are often considered less interpretable than traditional machine learning approaches. The internal decision-making process can be difficult to understand, as predictions result from multiple layers of learned parameters rather than explicitly defined rules or transparent feature representations. This reduced interpretability may pose challenges in regulated domains such as GDPR-compliant PII detection, where understanding and auditing model decisions can be important[28].

Hybrid

Every modeling approach has both strengths and limitations, and as a result, many practical PII detection systems adopt hybrid architectures that combine rule-based and machine learning components. Such hybrid models can improve overall performance by leveraging the precision of deterministic rules alongside the flexibility of statistical or neural models. However, this improvement often comes at the cost of increased system complexity[29].

The added complexity makes hybrid systems more difficult to maintain, deploy, and debug. When errors occur, it can be challenging to determine whether the issue originates from the rule-based component, the machine learning model, or their interaction. Furthermore, the components may produce conflicting predictions, requiring a predefined conflict-resolution strategy. Designing and validating such mechanisms introduces additional implementation and governance challenges[30][31].

Deep learning models offer certain advantages over classical machine learning approaches, particularly in their ability to learn feature representations automatically from data, reducing the need for manual feature engineering. However, contrary to common misconceptions, deep learning systems still require retraining when data distributions change or when new types of PII must be detected. Additionally, deep neural networks typically demand significantly greater computational resources, including higher memory usage and specialized hardware such as GPUs, which may increase operational costs[32].

2.4.2 State of the art in practice

State of the art models

There are several PII detection models available on the market, and many are hybrid architectures that combine rule-based methods with machine learning models. Examples include Microsoft Presidio, Google Cloud Sensitive Data Protection, and Amazon Macie.

Microsoft Presidio is an open-source framework that primarily combines rule-based pattern matching (e.g., regular expressions) with classical NLP models as spaCy-based NER. It does not include deep learning models by default, but it can be extended with transformer-based or other neural models. It is therefore best described as a hybrid system with optional deep learning components[33].

Google Cloud Sensitive Data Protection is a managed cloud service that integrates pattern matching with proprietary machine learning models. While Google does not publicly document its full architecture, it is most likely that deep learning-based NLP models are used for contextual entity recognition. The system is hybrid, combining deterministic detectors with ML and likely deep learning components[34].

Amazon Macie is an AWS-native service focused on detecting sensitive data in cloud storage. It combines rule-based detection with machine learning-based classification. Although Amazon does not disclose full technical details, Macie is described as using machine learning models, and it likely incorporates deep learning for certain classification tasks. It is therefore also considered a hybrid solution[35].

In practice, fully rule-based or purely classical machine learning front-end solutions without deep learning have become less common in commercial offerings. However, some organizations continue to rely on non-deep-learning approaches in internal or on-premise deployments. This is often motivated by considerations such as lower hardware requirements, reduced computational costs, greater interpretability, and increased operational stability. In regulated environments, transparency and deterministic behavior may be prioritized over marginal performance gains from deep neural models[29][30][32].

Model in practices

The model developed by Microsoft offers a wide range of functionalities, making it highly versatile in real-world applications. The system is designed to process unstructured text input, such as log files, documents, or user-generated content, and automatically identify and classify PII. Once detected, the PII elements are highlighted and presented in a structured format, typically in a table that specifies the entity type, name, email address, phone number and more, along with confidence scores. This structured output improves

transparency and allows users to review and validate the detections before further processing[33].

After identifying sensitive information, the system can modify the original string in multiple ways depending on the intended use case. For example, it can redact the information entirely, mask parts of it or apply pseudonymization techniques as shown in 2.3. One advanced option is the use of a deep learning-based approach, where detected PII is replaced with realistic but fictional data that mimics the format and semantic characteristics of the original information. For instance, a detected name may be substituted with another plausible name, and an address may be replaced with a syntactically valid but fabricated address.

This approach is particularly useful in logging and testing scenarios, where maintaining readability and contextual coherence is important. By generating synthetic yet realistic substitutes, the system preserves the overall structure and interpretability of the data while ensuring that no actual personal information is exposed. As a result, the model supports compliance with data protection regulations such as GDPR as mentioned in 2.2.2, while still enabling debugging, analytics, and system monitoring activities.

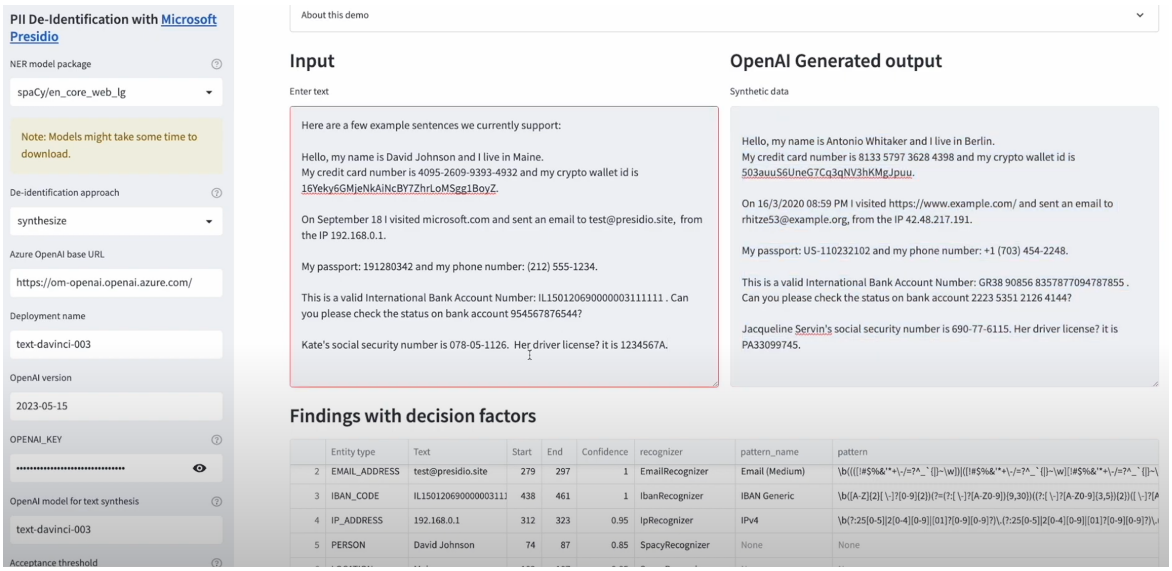

Figure 2.3: Screenshot of the user interface for Microsoft Presidio showing AI solution for anonymization of PII data[33]

Identifying companies that publicly disclose the use of a specific PII detection system is difficult, as security architectures and internal data protection are rarely shared in detail. Organizations generally avoid publishing such information to reduce potential security

risks. In the case of Microsoft Presidio, this challenge is increased by the fact that it is distributed as an open-source framework rather than a commercial off-shelf product.

As described in Atri[36], Presidio follows a modular architecture consisting of an Analyzer component for PII detection and an Anonymizer component for configurable data transformation fig2.4. This design enables organizations to integrate the framework at different stages of their data pipelines and adapt it to specific regulatory and technical requirements. However, because implementations vary across organizations, concrete company deployments are rarely documented publicly.

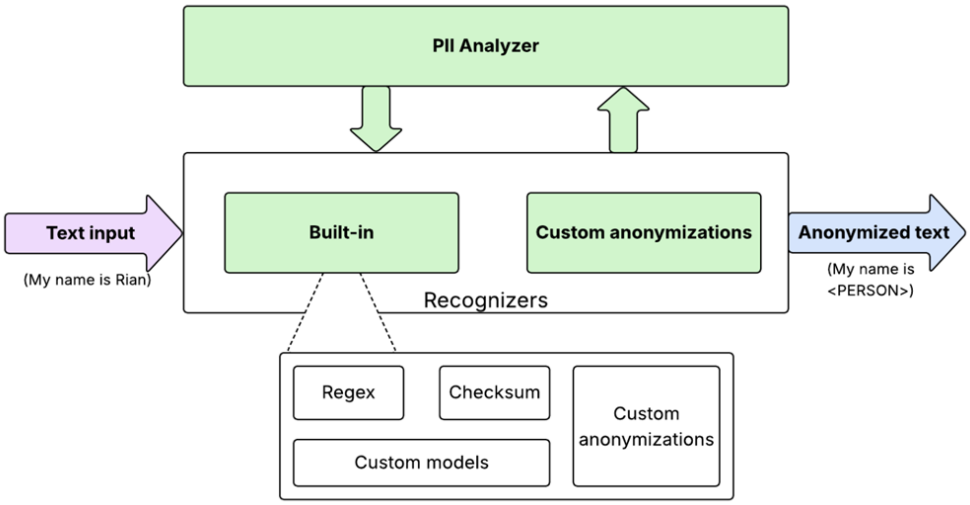

Figure 2.4: Architecture of A PII Detection and Anonymization Framework[36]

Summary

The different model types each have scenarios in which their strengths and weaknesses become evident. No single model has been shown to effectively solve all aspects of the PII detection problem. For this reason, a hybrid model without deep learning appears to be the most suitable solution, particularly since the group currently does not have the technical foundation to develop and train a deep-learning model.

The literature presented in this section supports this approach, as a hybrid of rule-based and classical machine learning models has shown promising results in practice. An approach without deep learning also provides better opportunities for implementing more interpretable models, where the decision-making is more transparent. In the context of PII detection, especially under laws such as GDPR, this level of interpretability is often

preferred, as it enables better documentation, validation, and reliability of the behavior of the system.

2.5 Purpose and Intended Users

This section combines the regulatory analysis presented earlier in sections 2.1.2 and 2.2.2. The objective is to clarify who the solution serves, where value is created, and why PII filtering is required for relevant stakeholders.
The necessity of a PII filtering system comes from the understanding that filtering is not an optional feature but a fundamental requirement in IT systems that process PII.[15]

2.5.1 Stakeholders

The primary users of a PII detection system are companies and organizations that handle digital personal data and are therefore the intended users of the final product. Thereby, key stakeholders are the data controllers (2.2), including roles such as Data Protection Officers and Security Officers [37]. These stakeholders interact directly with the system to ensure that sensitive information is identified and properly handled before it is logged or processed.

At the same time, customers or data subjects (2.2) are also stakeholders, but more indirectly. They benefit from the system because it protects their personal data, reduces the risk of privacy violations, and ensures that their information is handled in compliance with regulations such as the GDPR. Although they do not directly interact with the system, their trust and data privacy are directly supported by its implementation.
By focusing on both the companies that implement the system and the individuals whose data is protected, this section clarifies the key groups that get value from a PII detection solution.

2.5.2 Purpose and Business Value

The final product's purpose is to automatically detect and classify PII in log data, and for businesses, the system provides value by improving data and privacy practices. It ensures that PII is detected to prevent data breaches and maintain compliance with the GDPR, thereby reducing the risk of fines and the need for manual inspection of large log datasets, which can be time-consuming and error-prone. Thereby, an automated solution can also strengthen trust between the company and its customers. [3] [10]

2.5.3 Limitations of Existing PII Detection Approaches

Despite the approaches presented in the State of the Art section (2.4), there are still several challenges when different companies need to detect PII. Many existing solutions are

developed and maintained privately within companies, making it difficult for others to reproduce these approaches. Furthermore, log data can vary between companies due to differences in data storage, operating systems, and programming languages.
As a result, PII detection must be handled differently in a system-specific manner.[38].

2.5.4 Conclusion

Overall, automated PII detection can be considered a necessary component in modern IT systems that process personal data. However, due to variations in system architectures and log structures, such solutions must be adapted to specific contexts. This highlights both the importance and the complexity of implementing effective PII detection, thereby motivating further research into making effective automated PII identification solutions.

2.6 Final Problem Statement

How can a hybrid model combining rule-based methods and machine learning be developed to support data controllers in detecting and classifying sensitive PII in text, and how does the approach perform compared to other available methods?

Chapter 3

Requirement specification

3.1 Proof of concept

This product is developed as a proof of concept, meaning it is intended to demonstrate the practical application of the final intended use, rather than being a fully developed or production-ready system

3.2 Use Case

The primary use case concerns a system designed to detect and classify sensitive PII in customer service emails, including information such as names, addresses, email addresses, phone numbers, bank information, and social security numbers. To ensure privacy and secure handling of sensitive data, any identified PII must be removed, masked, or anonymized before being stored.

The system is developed as a technical solution intended for use by data controllers, who are responsible for ensuring the proper handling of personal data. While the system supports the identification and classification of sensitive PII, it remains the responsibility of the data controller to take appropriate actions based on the detected information, such as anonymization, deletion, or secure storage.

3.2.1 Scenario

1. The system receives a customer service email as input.
2. The system analyzes the text and detects potential sensitive PII using a combination of rule-based methods and machine learning techniques.

3. The detected PII is classified into predefined categories (e.g., names, email addresses, phone numbers, social security numbers).
4. The system outputs the detected and classified PII, which is forwarded to the data controller for further handling and decision-making.
5. The data controller is responsible for taking appropriate actions based on the output, such as anonymization, deletion, or secure storage of the data.
6. The original text is not stored after processing, ensuring that sensitive information is not retained unnecessarily.

3.3 Definitions and assumptions

3.3.1 Model Transparency

In the context of PII detection, model interpretability is an important consideration. White-box Machine Learning models are characterized by transparent decision processes, where it is possible to see how predictions are generated and which input variables influence the model's output. This transparency makes it possible to validate model behaviour and identify potential errors in predictions. Such properties are particularly relevant in systems that process sensitive personal data, where understanding and auditing model decisions may be necessary for compliance and system reliability [39].

In this project, the aim is therefore to favor an interpretable or semi white-box approach, where the classification process can be analysed and inspected. By using models such as Support Vector Machine (SVM) combined with explicit feature representations, it becomes possible to examine how textual features contribute to the classification of PII. This provides a higher degree of transparency compared to purely black-box models, such as deep neural networks.

3.4 Functional and non-functional Requirements

This project's requirements are divided into two categories, Functional Requirements and Non-Functional Requirements. Functional Requirements describe what the system should do and the actions it must perform to work. Non-Functional Requirements describe how the system should perform the Functional Requirements and ensure that user needs are fulfilled. [40]

3.4.1 Functional Requirements

ID	Requirement	Traceability
FR-1	The system must detect and identify sensitive PII in textual input data.	3.2.1
FR-2	The system must classify detected PII into predefined categories (e.g., name, email, phone number, social security number).	3.2.1
FR-3	The system must accept and process unstructured text input (e.g., customer service emails).	3.2.1
FR-4	The system must support sequential text processing for context-aware classification.	3.2.1
FR-5	The system must combine rule-based methods and machine learning techniques in a hybrid approach.	3.2
FR-6	The system must be able to train on annotated datasets.	x
FR-7	The system must output detected and classified PII in a structured format.	3.2.1
FR-8	The system must provide results that can be forwarded to a data controller for further handling.	3.2.1
FR-9	The system must support evaluation of model performance.	x

Table 3.1: Functional requirements

3.4.2 Non-Functional Requirements

ID	Requirement	Traceability
NFR-1	The system must be interpretable and function as a white-box or explainable model.	x
NFR-2	The system must achieve precision on at least x%	x
NFR-3	The system must achieve recall on at least x%	x
NFR-4	The system must process input data within a maximum of 100 milliseconds	x
NFR-5	The system must not store original sensitive text after processing.	3.2.1
NFR-6	The system must ensure secure handling of sensitive data during processing.	3.2.1

Table 3.2: Non-functional requirements

3.5 Delimitation

As shown in Table 3.2, the requirement NFR-3 prioritizes minimizing false negatives to ensure that as much PII as possible is detected. In this context, false negatives are considered more critical than false positives, as undetected PII may lead to data breaches. Consequently, accuracy would be lower than in a balanced system, and for this project, it is acceptable as ensuring high recall is more important.

Chapter 4

Methods

4.1 Data Science Lifecycle

4.2 Dataprocessering databalancering

4.3 Feature engineering

4.4 Validation setup

4.5 Supervised Learning

Supervised learning is a learning paradigm that uses labeled data. The labels represent the desired outputs and are called "ground truth" for the training data provided to the algorithm, making it a common approach for classification and regression tasks. Therefore, when using supervised learning, the goal is to create a model that can predict correct outputs on new, unseen data. Furthermore, the term “ground truth” is used because these labels represent verified, real-world outcomes and help provide a standard for evaluating the trained model. [41] [42]

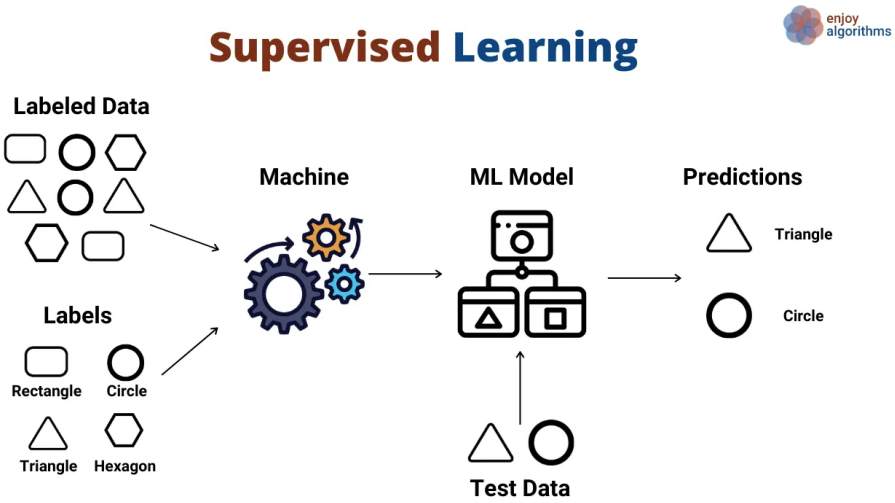

Figure 4.1: Illustration of Supervised Learning [43]

Figure 4.1 visualizes how Supervised Learning works .....

4.6 Rule-based systems

4.6.1 Regex

Regular Expressions, also called Regex, are a rule-based system used to describe and search for patterns in strings. Regex allows predefined patterns to be specified to filter or detect specific character sequences. It can be as simple as searching for a comma, or as complex as searching for an email address in a text.
Therefore, it can be used to build a system that classifies whether a text contains PII or not by detecting patterns such as email addresses, CPR numbers, or phone numbers.
As an example, emails can be found by using the following rule:

```
[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}
```

Explanation of the pattern:

Regex Part	Explanation
`[a-zA-Z0-9._%+-]+`	Matches the username part of an email. It allows: • letters from a to z (lowercase) and A to Z (uppercase) • digits 0–9 • dot (.), underscore (_), percent sign (%), plus (+), and hyphen (-) The '+' means one or more of these characters.
`@`	Matches the "@" symbol, which separates the username from the domain.
`[a-zA-Z0-9.-]+`	Matches the domain name. It allows: • letters from a to z (lowercase) and A to Z (uppercase) • digits 0–9 • dot (.) • hyphen (-) The '+' means one or more of these characters, so the domain must have at least one character.
`\.[a-zA-Z]{2,}`	Matches the top-level domain, such as '.com' or '.dk'. The backslash escapes the dot so it is taken literally. '[a-zA-Z]2,' means the domain must contain at least 2 letters, uppercase or lowercase.

Table 4.1: Detailed explanation of each part of the email regex pattern.

Using this regex (table 4.1, it can extract all emails from the dataset. This kind of rule-based system does not require any training and works immediately once the rule is defined. These regex rules must be manually created, which can become challenging when dealing with large or complex text datasets. Furthermore, because the system is not trained, regex cannot understand the context of text and only matches predefined patterns. This limitation makes it less effective for tasks requiring complex language understanding compared to Machine Learning (ML) models [44] [45].

4.7 Machine Learning

4.7.1 Conditional Random Fields

This section is largely based on the tutorial by Sutton and McCallum[46], which provides a detailed introduction to CRF and their application to structured prediction tasks.

CRF are a class of Machine Learning(ML) models designed for problems where output variables are not independent of each other. In many domains such as natural language processing, bioinformatics, and computer vision, the elements that must be classified are

closely related. In such cases it is beneficial to model dependencies between output variables instead of classifying each observation independently. Relational data typically exhibits two important characteristics. First, statistical dependencies exist between the entities that must be modeled. Second, each entity often has a rich set of observable features that can assist classification. A common example occurs in text processing, where the meaning or classification of a word often depends on its surrounding context.

Graphical models provide a natural framework for representing such dependencies. In general, graphical models describe probability distributions over sets of random variables using a graph structure in which nodes represent variables and edges represent dependencies between them. Traditionally, many models attempt to represent the joint probability distribution between inputs and outputs, written as

$$p(y, x) \tag{4.1}$$

where x represents the observed input variables and y represents the output variables that must be predicted. However, modeling the joint distribution can be difficult because it requires learning the distribution of the input data $p(x)$. In many practical applications, input features are highly dependent on each other, making this distribution complex and difficult to estimate. If these dependencies are ignored, the model may lose important information, while explicitly modeling them can make the model computationally intractable.

CRF address this problem by directly modeling the conditional probability distribution

$$p(y|x) \tag{4.2}$$

which represents the probability of an output configuration given the observed input. By focusing only on the conditional distribution, CRF avoids the need to explicitly model the distribution of input variables. This allows the model to incorporate many overlapping and complex features without requiring independence assumptions among them. As a result, CRF are particularly well suited for structured prediction tasks in natural language processing, where contextual information and feature overlap are common.

Graphical models used for classification are often divided into two main categories, generative models and discriminative models. Generative models attempt to describe how data is generated by modeling the joint probability distribution $p(y, x)$. A well-known example is the Naive Bayes classifier, which assumes that input features are conditionally independent given the class label.

Discriminative models model the conditional distribution $p(y|x)$ directly. Logistic regression is a classical example of such a model. In logistic regression, the probability of a class is expressed as an exponential function of weighted feature functions:

$$p(y|x) = \frac{1}{Z(x)} \exp\left(\sum_k \lambda_k f_k(y, x)\right) \tag{4.3}$$

where $f_k(y, x)$ are feature functions describing properties of the input and output, λ_k are learned weights, and $Z(x)$ is a normalization term ensuring that probabilities sum to one. Conditional Random Fields extend this discriminative approach to structured outputs such as sequences or graphs, where the output variables are not independent.

The most commonly used form of CRF is the linear-chain Conditional Random Field, which is specifically designed for sequence labeling tasks. In sequence labeling problems, the input consists of a sequence

$$x = (x_1, x_2, ..., x_T) \tag{4.4}$$

and the objective is to predict a corresponding sequence of labels

$$y = (y_1, y_2, ..., y_T). \tag{4.5}$$

A key observation in many sequential problems is that neighboring labels are dependent. For example, in named entity recognition, if a word is labeled as the beginning of a location name, the following word is likely to belong to the same entity. Modeling such dependencies improves classification accuracy because the prediction of one label influences the prediction of neighboring labels.

Linear-chain CRF represent these dependencies by connecting the output variables in a chain structure. Each label depends on the input data as well as the previous label in the sequence. The conditional probability of a label sequence given an input sequence is defined as

$$p(y|x) = \frac{1}{Z(x)} \exp\left(\sum_{t=1}^{T} \sum_k \lambda_k f_k(y_t, y_{t-1}, x, t)\right) \tag{4.6}$$

where $f_k(y_t, y_{t-1}, x, t)$ are feature functions describing both the current input token and the transition between neighboring labels, and λ_k are the parameters of the model. The normalization constant $Z(x)$, known as the partition function, ensures that the probabilities sum to one and is defined as

$$Z(x) = \sum_y \exp\left(\sum_{t=1}^{T} \sum_k \lambda_k f_k(y_t, y_{t-1}, x, t)\right). \tag{4.7}$$

Feature functions capture relevant properties of the data. In natural language processing tasks they often describe properties such as whether a word starts with a capital letter,

whether it contains digits, which words appear before or after it, or whether the word appears in a predefined dictionary. Because CRF do not require independence assumptions among features, many different types of features can be used simultaneously.

Training a CRF involves estimating the parameters λ_k from labeled training data. Suppose a dataset contains input-output pairs

$$D = \{(x^{(i)}, y^{(i)})\}_{i=1}^{N}. \tag{4.8}$$

The parameters are typically learned by maximizing the conditional log-likelihood

$$l(\theta) = \sum_{i=1}^{N} \log p(y^{(i)}|x^{(i)}). \tag{4.9}$$

To prevent overfitting, a regularization term is usually added, resulting in

$$l(\theta) = \sum_{i=1}^{N} \log p(y^{(i)}|x^{(i)}) - \frac{1}{2\sigma^2} \sum_{k} \lambda_k^2. \tag{4.10}$$

This regularization corresponds to assuming a Gaussian prior over the parameters. The objective function is convex for fully observed CRF, meaning that optimization algorithms are guaranteed to find a global optimum. In practice, numerical optimization methods such as Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) or conjugate gradient are commonly used to perform this optimization efficiently.

Once the model has been trained, it can be used to perform inference on unseen data. Two common inference tasks arise. The first is computing the partition function and marginal probabilities needed during training. The second is predicting the most probable sequence of labels for a given input sequence. Both tasks can be solved efficiently using dynamic programming algorithms.

The forward algorithm computes intermediate probabilities that summarize all possible partial label sequences up to a given position:

$$\alpha_t(j) = \sum_{i} \alpha_{t-1}(i) \Psi_t(j, i, x_t) \tag{4.11}$$

where

$$\Psi_t(y_t, y_{t-1}, x_t) = \exp\left(\sum_{k} \lambda_k f_k(y_t, y_{t-1}, x_t)\right). \tag{4.12}$$

To predict the most probable label sequence, the Viterbi algorithm is used. It finds

$$y^* = \arg\max_{y} p(y|x) \tag{4.13}$$

by recursively computing

$$\delta_t(j) = \max_i \left[\delta_{t-1}(i)\Psi_t(j, i, x_t)\right]. \tag{4.14}$$

These algorithms allow inference in linear-chain CRF to be performed in

$$O(T|S|^2) \tag{4.15}$$

time, where T is the sequence length and $|S|$ is the number of possible labels.

Linear-chain CRF are a special case of the broader CRF framework. In the general formulation, output variables may form arbitrary graph structures instead of simple chains. A general CRF can be written as

$$p(y|x) = \frac{1}{Z(x)} \prod_{A \in G} \exp\left(\sum_k \lambda_{A,k} f_{A,k}(y_A, x_A)\right) \tag{4.16}$$

where G represents the set of cliques in the graphical model. More complex variants such as skip-chain CRF add additional edges between distant nodes in a sequence to capture long-range dependencies. Although these models can represent more complex relationships, they are computationally more expensive and often require approximate inference algorithms.

In many practical applications, particularly in text processing tasks, the simpler linear-chain CRF structure is sufficient because most relevant dependencies occur between neighboring elements in the sequence. As a result, linear-chain CRF provide a practical balance between expressive power and computational efficiency. They remain one of the most widely used models for structured prediction tasks where sequential dependencies play an important role.

4.7.2 SVM

A SVM is a supervised machine learning algorithm commonly used for classification tasks, although it can also be applied to regression problems. The main objective of an SVM is to identify the optimal boundary that separates different classes within a dataset. This boundary is known as a hyperplane, which divides the data points in a multidimensional feature space[47].

$$w^T x + b = 0 \tag{4.17}$$

where w represents the weight vector, x is the feature vector, and b is the bias term. This equation defines the boundary that separates the classes in the feature space [48]. Maximizing this margin improves the model's ability to generalize to unseen data [48]. Rather than simply separating the data, an SVM attempts to find the hyperplane that maximizes

the distance between the boundary and the nearest data points from each class. This distance is referred to as the margin. By maximizing the margin, the model improves its ability to generalize to unseen data and reduces the risk of overfitting[47]. The margin can be expressed as:

$$Margin = \frac{2}{||w||} \tag{4.18}$$

The data points that lie closest to the decision boundary are called support vectors, and these points determine the position and orientation of the hyperplane[47]. Because the model focuses on the most informative samples, SVM can perform well even when working with complex or high-dimensional datasets.

This characteristic makes SVM particularly suitable for text classification tasks, such as the detection of Personally Identifiable Information PII. In PII detection systems, textual data is typically transformed into numerical feature representations using techniques such as TF-IDF[49], where words or tokens are represented as vectors in a high-dimensional space. SVM performs well in such environments because it can efficiently separate classes in large feature spaces and identify patterns associated with sensitive information.

In the context of PII detection, the model learns to distinguish between PII and non-PII text segments[49]. For example, it can learn patterns associated with personal names, addresses, phone numbers, or other identifying attributes based on the statistical distribution of words and tokens in the training data. Because SVM focuses on defining a clear decision boundary, it can serve as an effective baseline model for identifying sensitive information in textual data streams.

Another advantage of SVM is its relatively low computational complexity compared to deep learning models, which can be beneficial in systems with strict latency requirements. This makes SVM a relevant candidate for environments where PII must be detected quickly, such as real-time data processing pipelines.

However, while SVM performs well for many classification tasks, it has limitations when dealing with complex linguistic context. Unlike deep learning models such as recurrent neural networks or transformer-based architectures, SVM does not inherently capture sequential relationships between words. As a result, more advanced neural models may achieve higher accuracy when contextual understanding of language is required.

Overall, SVM provides a robust and computationally efficient approach for PII detection, particularly in high-dimensional text classification problems. It can therefore serve as a strong baseline model for identifying sensitive information before data is further processed, stored, or logged in a system.

TF-IDF

Term Frequency-Inverse Document Frequency (TF-IDF) is a statistical method used to evaluate the importance of a given term in a collection of documents. TF-IDF is commonly used for keyword extraction and document classification. TF-IDF is composed of two elements: Term Frequency (TF) & Inverse Document Frequency (IDF).
Term Frequency (TF) is the measurement of how often a given word appears in a document, words with higher frequency are generally considered more important in describing the document's content. The TF value is calculated using the following formula.

$$TF(t,d) = \frac{Total\ number\ of\ times\ term\ t\ appears\ in\ document\ d}{Total\ number\ of\ terms\ in\ document\ d}$$

The second component is Inverse Document Frequency (IDF). This measures how rare or unique a term is across multiple documents, The purpose of IDF is to reduce the weight of commonly occurring terms while increasing the importance of terms that appear less often across multiple documents. the IDF value is calculated with this formula.

$$IDF(t,D) = \log \frac{Total\ number\ of\ documents\ D}{Total\ number\ of\ documents\ containing\ term\ t}$$

By combining both TF & IDF, a method is created that highlights terms that are commonly used in a given document and unique across multiple documents. This makes TF-IDF an effective method for text classification. In combination with other methods TF-IDF can be used for PII detection in documents. [50] [51]

4.8 Hyperparameter optimization

Chapter 5

Implementation

Chapter 6

Test/Results

Chapter 7

Discussion

7.1 Ethics

Chapter 8

Conclusion

Here we conclude things and the chapter number is 8.

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Appendix A

Appendix

Here is the appendix